

Vulnerability Intelligence Platform

Course of Big Data Technologies - 2025/2026

Group 5

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TEAM, ROLES AND COLLABORATION



TEAM AND ROLES

Cristina Morillo Leal: BSc in Business Administration and Management and BEng in Computer Engineering

- **Responsibilities:** Infrastructure, orchestration, data ingestion and construction of the master vulnerability dataset.

Francesco Leoni: MSc in Work and Organizational Psychology (LM-51)

- **Responsibilities:** Data quality, vulnerability scoring, clustering and cluster-aware risk refinement.

Thuy Trang Vo: BSc in Information, Communication and Technology

- **Responsibilities:** Simulation and results layers, remediation action, DuckDB query interface, app dashboards.

We also cross-reviewed each other's work to catch errors early and keep the final output consistent.



COLLABORATION TOOLS

Github for version control

Whatsapp for immediate communication

Google Meet for meetings

Google Drive for drafts and shared documents

ABSTRACT

AIM:

Our project delivers a scalable vulnerability intelligence platform that helps security teams prioritize daily remediation efforts under limited capacity. The system ingests public vulnerability feeds, enriches them with risk signals, and converts them into an actionable, ranked patch plan.



KEY OBJECTIVES

- Scalable ingestion and storage of public vulnerability data (NVD, CISA KEV, EPSS).
- Data cleaning, normalization, and master dataset construction.
- Priority scoring and cluster-aware risk analysis.
- Capacity-based remediation planning and interactive exploration through Streamlit.



MAIN RESULTS

- Implemented an end-to-end data pipeline from ingestion to serving with Spark and Docker.
- Built a gold-layer master vulnerability table by joining NVD, KEV, and EPSS on cve_id.
- Delivered interpretable priority scores, vulnerability clusters, and remediation recommendations.
- Created an interactive dashboard for exploration, prioritization, and decision support.

PROBLEM STATEMENT AND MOTIVATION



THE PROBLEM

- **Vulnerability data is fragmented:** NVD, KEV, and EPSS provide different signals that are not directly comparable.
- **Prioritization is slow and error-prone:** Security teams cannot manually rank thousands of CVEs with enough consistency.
- **Capacity is limited:** Only a small fraction of vulnerabilities can be patched each day, so choosing what to fix first matters.
- **Risk signals are noisy without context:** A high CVSS score alone does not always mean a vulnerability should be treated first.



MOTIVATION

- It helps security teams focus on the vulnerabilities that matter most.
- It turns public vulnerability feeds into a practical remediation plan.
- It supports faster and more consistent decision-making under limited capacity.
- It reduces the chance that high-risk issues stay unpatched for too long.



WHY IS BIG DATA PROBLEM

- **Volume:** NVD covers roughly 280K CVEs and grows by about 150-250 vulnerabilities per day; EPSS is similar in size, while KEV has around 1.5K entries.
- **Variety:** NVD is in JSON, while KEV and EPSS are in CSV.
- **Velocity:** NVD modified feeds update every 2 hours, KEV updates with new entries or changes during business hours, and EPSS is a daily snapshot.
- **Complexity:** The data has to be deduplicated, normalized, enriched, clustered, and ranked before it becomes actionable.

DATA EXPLORATION INSIGHTS

OVERVIEW TABLE

Layer	Dataset / Table	Key Fields	Purpose
Raw	NVD	cve_id, published, last_modified, description, cwes, cvss_score, cpe_vendors, cpe_products, reference_count	Authoritative CVE catalog in JSON; base dataset
Raw	CISA KEV	cve_id, vendor_project, product, kev_date_added, required_action, is_ransomware	Known exploited vulnerabilities in CSV
Raw	EPSS	cve_id, epss_score, epss_percentile, score_date	Daily exploitability snapshot in CSV
Silver	Normalized feeds	cleaned schemas, deduplicated CVEs, partitioned Parquet/Delta	Standardized data ready for joins
Gold	master_vulnerabilities	is_kev, epss_score, priority_score, cluster_id, primary_vendor, primary_product	Analytics-ready master table for scoring and clustering
Gold	remediation outputs	strategy comparison, cluster risk summary, remediation actions	Final decision-support outputs

DATA HIGHLIGHTS OF A SAMPLE RUN

TOTAL CVEs: 280,047 vulnerabilities in the snapshot

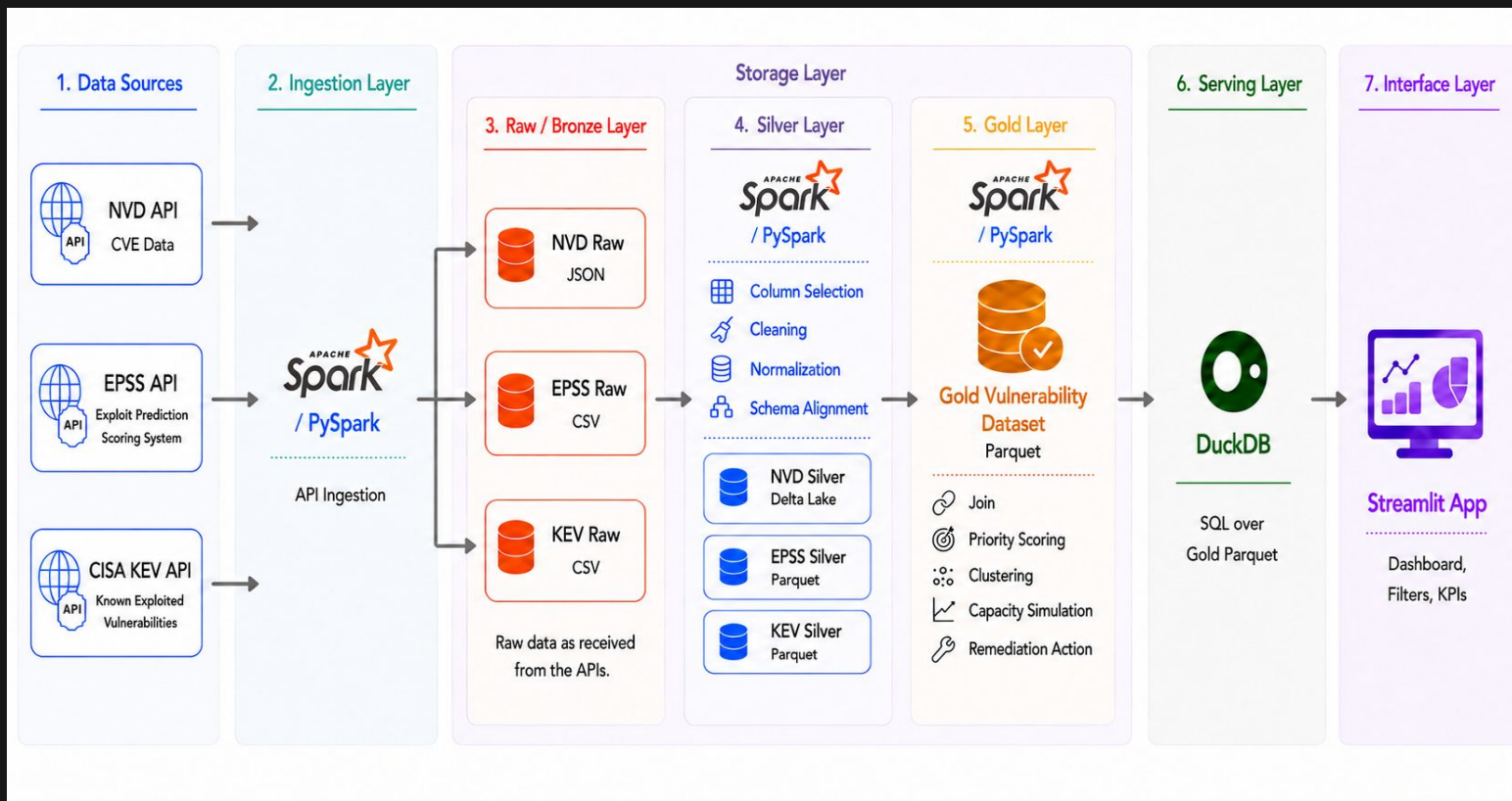
CISA KEV: 1,451 known exploited vulnerabilities

% CRITICAL: 6.0% of the catalog

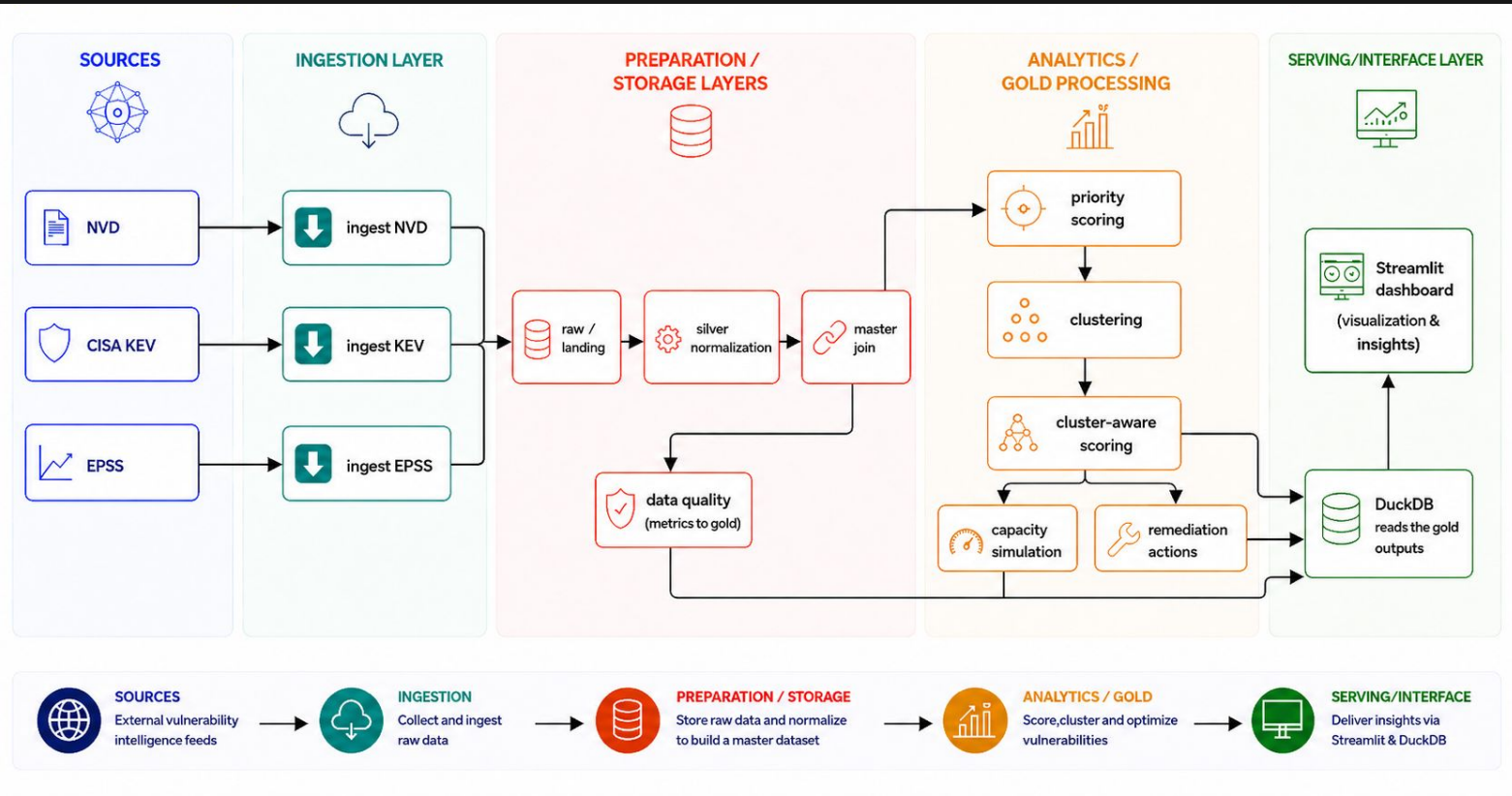
EPSS > 0.7: 3,049 high-exploitability CVEs

FINAL OUTPUTS: master table, priority scores, clusters, and remediation plan

ARCHITECTURE



PIPELINE: HOW DATA FLOWS



TECHNOLOGIES AND JUSTIFICATION



Spark

Aim: Scalable data processing

Role: Ingests, joins, scores, clusters, and writes partitioned outputs at scale.



Delta Lake

Aim: Transactional storage

Role: Supports the NVD silver layer with dynamic overwrite and update-friendly snapshots.



Parquet

Aim: Efficient columnar storage

Role: Ingests, joins, scores, clusters, and writes partitioned outputs at scale.



DuckDB

Aim: Fast read-only querying

Role: Reads the gold layer efficiently for the dashboard without a separate database server.



Streamlit

Aim: Interactive exploration

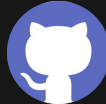
Role: Provides the dashboard for vulnerability analysis, filtering, and remediation planning.



Docker Compose

Aim: Reproducible deployment

Role: Packages the app and pipeline so the project runs consistently on any machine.



GitHub

Aim: Version control

Role: Tracks code changes and supports teamwork across the whole project.



Why not Kafka?

Kafka was not used because the project is batch-based, not streaming-based.

IMPLEMENTATION AND CODE REPOSITORY

CODE STRUCTURE

```
├── README.md           ─ this file
├── requirements.txt     ─ Python package dependencies
├── Dockerfile          ─ image definition for app + pipeline
├── docker-compose.yml  ─ local app/pipeline service setup
├── docker-entrypoint.sh ─ container command router
├── .dockerignore       ─ excludes data, caches, and build junk
├── .gitignore          ─ excludes generated files and local-only artifacts
├── team_contributions/ ─ individual team member contribution notes
├── src/
│   ├── config.py       ─ paths, defaults, Spark session factory
│   ├── ingestion/
│   │   ├── ingest_nvd.py      ─ full NVD yearly ingest
│   │   ├── ingest_nvd_modified.py ─ modified-feed incremental upsert
│   │   ├── ingest_kev.py      ─ KEV snapshot ingest
│   │   └── ingest_epss.py     ─ daily EPSS snapshots
│   ├── processing/
│   │   ├── join_master.py     ─ master assembly
│   │   └── data_quality.py    ─ snapshot metrics
│   ├── scoring/
│   │   ├── priority_scoring.py ─ base score
│   │   └── cluster_aware_scoring.py ─ cluster-adjusted score
│   ├── clustering/
│   │   └── clustering.py      ─ vulnerability clustering
│   ├── optimization/
│   │   ├── capacity_simulation.py ─ strategy simulation
│   │   └── remediation_actions.py ─ ranked patch plan
│   ├── pipeline/
│   │   └── daily_pipeline.py   ─ end-to-end orchestrator
│   └── utils/
│       ├── http.py           ─ download helpers + retries
│       └── duckdb_helpers.py  ─ serving-layer SQL helpers
├── app/
│   └── streamlit_app.py      ─ Streamlit UI
├── tests/
│   └── test_*.py            ─ full pytest suite
├── data/
│   ├── raw/
│   ├── silver/
│   └── gold/
└──
```

HOW TO RUN THE PROJECT

PREREQUISITES:

- Docker (and Docker Compose) installed on your machine
- 8GB RAM
- ≥8 CPU Cores

LAUNCHING:

- Clone the repo: **git clone**
<https://github.com/crismorillo01/Big-Data-Technologies-Project.git>
- Open Docker Desktop on your machine
- Build and launch: once inside the Big-Data-Technologies-Project directory, run **docker compose up --build -d**
- Wait for services to initialize
- Ingested data will be processed and available in the dashboard.
- Open the dashboard at **<http://localhost:8501>**

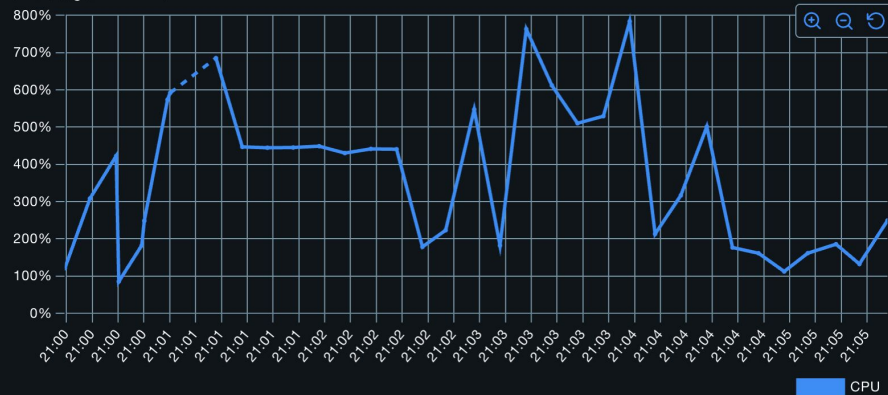
Notes: See full setup and usage in README.md on GitHub!

RESULTS AND PERFORMANCE

CPU USAGE:

- The CPU usage is highly variable during the execution, with several peaks close to the maximum available capacity.
- This suggests that the system performs computationally intensive tasks at specific moments.
- Despite the spikes, the CPU usage stays within the container limit, so the execution remains stable.

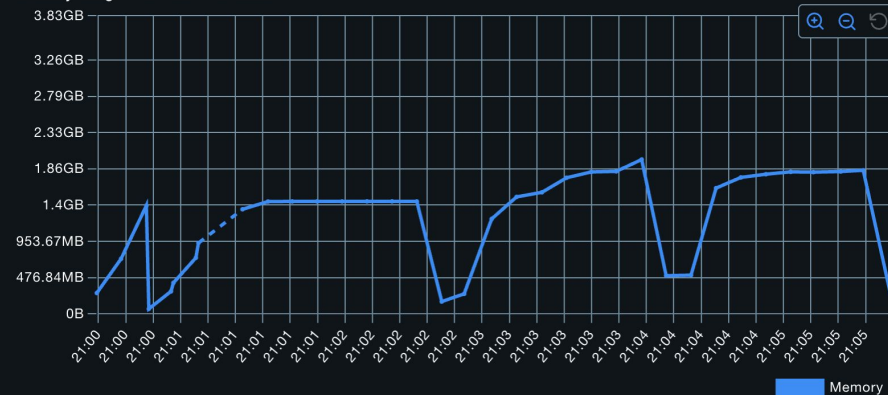
CPU usage: 249.47%



MEMORY USAGE:

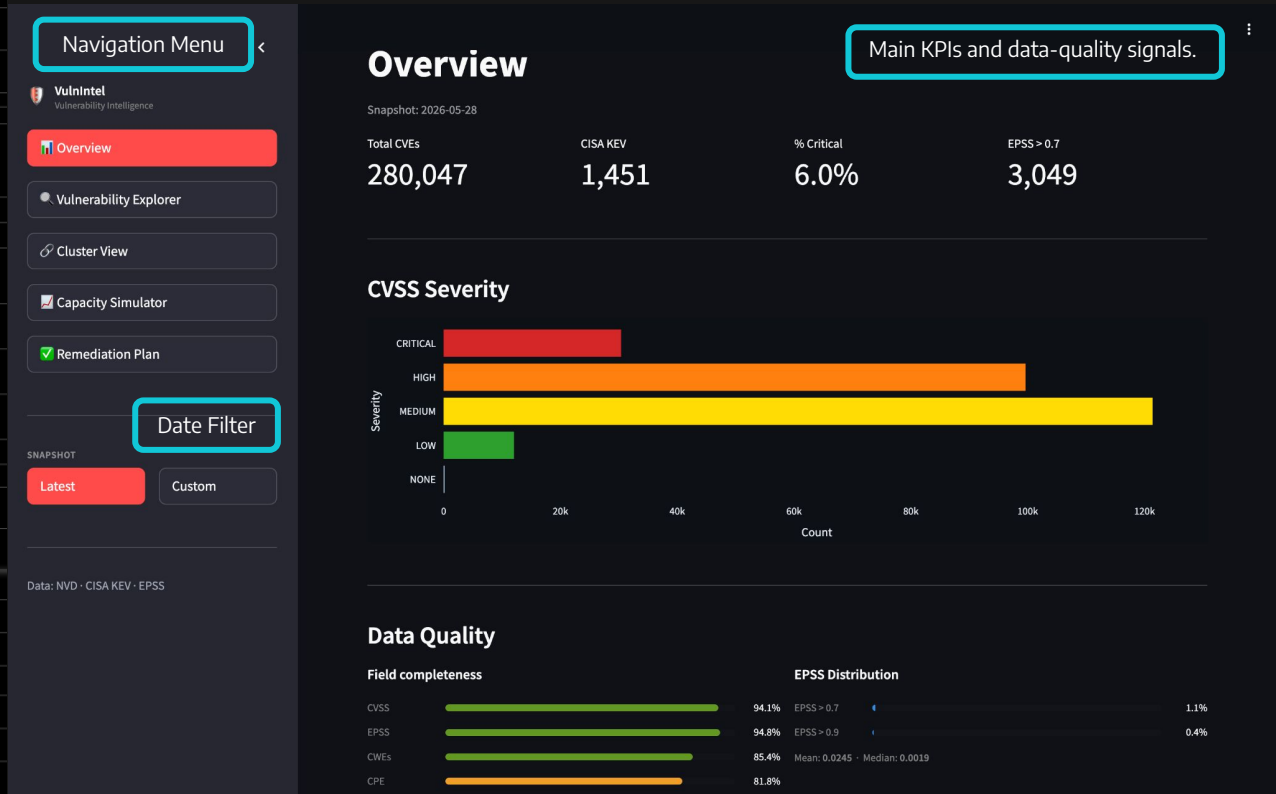
- Memory usage remains much more stable than CPU usage.
- After the initial ramp-up, memory consumption stays around 1.5–1.9 GB for most of the execution.
- Since the memory limit is 3.83 GB, the system has enough available memory and does not show signs of memory saturation.

Memory usage: 267.3MB / 3.83GB



DASHBOARD

The dashboard lets users explore vulnerability data, inspect risk signals, and review remediation priorities through an interactive interface.



Filter by available snapshot date to update the dashboard.



DASHBOARD / RESULTS

Show top

Top-N selector

10

25

50

100

Remediation plan by
vendor/product

Export CSV

Export the table as CSV

#	Vendor	Product	CVEs	KEV	Max priority	Mean EPSS	Action score
1	unknown	unknown	51,031	0	Critical	0.01	1692.42
2	linux	linux_kernel	11,953	9	Critical	0.00	495.57
3	google	android	7,642	17	Critical	0.00	382.58
4	microsoft	windows_10	2,876	2	Critical	0.08	249.11
5	google	chrome	3,056	67	Critical	0.02	236.49
6	mozilla	firefox	1,698	9	Critical	0.01	137.17
7	microsoft	windows_10_1507	1,206	130	Critical	0.07	117.46

This table ranks vendor/product groups by action score. It combines the number of CVEs, KEV count, max priority, and mean EPSS to show where remediation effort should go first.

GO

google / android

7,642 CVEs · 17 in KEV · Critical

Total CVEs

7,642

KEV count

17

Max EPSS

0.87

Action score

382.58

Top CVEs

CVE-2019-2215

CVE-2020-0041

CVE-2023-20963

CVE-2025-48572

CVE-2025-48633

Selecting a row updates the summary below with the chosen group's main metrics and top CVEs, so you can inspect the remediation target in more detail.

DASHBOARD / RESULTS

Remediation plan by capacity strategy

Strategy

Top priority

KEV COVERAGE

3%

Share of KEV CVEs captured by the strategy

EPSS MITIGATED

45.73

Cumulative EPSS score selected

CLUSTER DIVERSITY

0.93

Entropy-style spread across clusters

SELECTED CVEs

50

Rows selected under the current capacity

Export CSV

#	CVE	Priority band	EPSS	KEV	CVSS	Cluster	Vendor	Product	Priority score
1	CVE-2023-20198	Critical	0.94	KEV	10.0	5	rockwellautomation	allen-bradley_stratix_5200_firmware	
2	CVE-2022-20699	Critical	0.89	KEV	10.0	5	cisco	rv340_firmware	
	CVE-								

CV CVE-2023-20198
rockwellautomation - allen-bradley_stratix_5200_firmware - Critical

Priority score

1.1

CVSS score

10.0

CRITICAL

EPSS score

0.940

Published

2023-10-16

Critical

CISA KEY

Cisco is providing an update for the ongoing investigation into observed exploitation of the web UI feature in Cisco IOS XE Software. We are updating the list of fixed releases and adding the Software Checker. Our investigation has determined that the actors exploited two previously unknown issues. The attacker first exploited CVE-2023-20198 to gain initial access and issued a privilege 15 command to create a local user and password combination. This allowed the user to log in with normal user access. The attacker then exploited another component of the web UI feature, leveraging the new local user to elevate privilege to root and write the implant to the file system. Cisco has assigned CVE-2023-20273 to this issue. CVE-2023-20198 has been assigned a CVSS Score of 10.0. CVE-2023-20273 has been assigned a CVSS Score of 7.2. Both of these CVEs are being tracked by CSCwh87343.

Required action: Verify that instances of Cisco IOS XE Web UI are in compliance with BOD 23-02 and apply mitigations per vendor instructions. For affected products (Cisco IOS XE Web UI exposed to the internet or to untrusted networks), follow vendor instructions to determine if a system may have been compromised and immediately report positive findings to CISA.

Strategy

Top priority

Top priority

High EPSS

Cluster based

KEV first

Hybrid

These key cards summarize how the selected strategy performs: KEV coverage, EPSS mitigated, cluster diversity, and selected CVEs.

The table ranks CVEs under the selected strategy. Clicking a row updates the panel below with the vulnerability details, scores, tags, and required action.

Top priority: Selects the CVEs with the highest final priority scores.

High EPSS: Selects the CVEs with the highest exploitability risk.

Cluster based: Distributes the selection across clusters to avoid concentrating on one risk pattern.

KEV first: Prioritizes known exploited vulnerabilities before filling the remaining capacity.

Hybrid: Combines KEV-first and cluster-based selection to balance urgency and diversity.

LIMITATIONS, LESSONS AND NEXT STEPS

LIMITATIONS

- The NVD modified feed supports incremental updates, but missed runs may require a full 12-year refresh.
- The dashboard depends on data generated by running the pipeline first.
- Some outputs rely on available NVD, KEV, and EPSS fields, so missing values can reduce detail.
- The scoring and clustering logic is interpretable, but still heuristic and not a ground-truth risk model.

NEXT STEPS

- Improve the dashboard with more comparison views and stronger visual summaries.
- Explore richer remediation logic and strategy tuning.
- Refine the scoring model with additional signals and sensitivity checks.
- Add trend analysis across multiple snapshots to track how vulnerability risk evolves over time.

LESSONS

- API retrieval and setup.
- Setup of docker compose containers.
- Dashboard configuration.
- Cross-reviewing the code helped catch inconsistencies early.
- A clear data model made the pipeline easier to extend and debug.

REFERENCES, ACKNOWLEDGEMENTS AND GENAI DISCLOSURE

GENAI USAGE

- **Codex** and **Claude** helped with project structure and code development.
- **ChatGPT** helped with text drafting and PPT diagram generation.

ACKNOWLEDGEMENTS

We would like to thank our families for making it possible for us to be here today.

DATASET LINKS

NVD:

<https://nvd.nist.gov/vuln/Data-Feeds>

KEV:

<https://www.cisa.gov/known-exploited-vulnerabilities-catalog>

EPPS:

https://www.first.org/epss/data_stats.html

THANK YOU